

Keuntae Kim

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Summary

PhD Candidate at Hanyang University working on two threads. **Interpretability & Safety:** white-box use of model internals — attention, entropy, confidence dynamics — to detect hallucination, assess reasoning quality, and steer decoding without retraining. **LLM / VLM (Video-LLM) Post-training:** RLVR and policy optimization, agentic RL for stateful agents, multimodal alignment, and continual learning, across LLMs and Video-LLMs. This has produced **5 first-author papers** at **NeurIPS 2025 / CVPR 2026 / ICML 2026 / EMNLP 2026 / NeurIPS 2026** and **5 first-author submissions** under review at ICLR 2027 and Nature. I also own the **distributed training stack** end-to-end, training LLMs and VLMs on **up to 32 H100s (4 nodes × 8)** across Samsung SDS Cloud, NHN Cloud (B200), and AWS (H200). **Researcher** on two Korean national AI projects (NRF, IITP); reviewer for NeurIPS, ICLR, ICML, CVPR, ECCV, and ACL ARR.

Experiences

Large-Scale & Multi-Node Training Experience

Samsung SDS Cloud / NHN Cloud / AWS

Hands-on owner of end-to-end distributed training for LLMs and VLMs on three production GPU clouds — cluster bring-up, NCCL / RDMA validation, parallelism strategy, and fault-tolerant multi-day runs.

Platform	Accelerator	Topology	Scale
Samsung SDS Cloud	NVIDIA H100 SXM (80 GB)	4 nodes × 8 GPUs	32 GPUs — largest configuration; NVLink/NVSwitch intra-node, RDMA inter-node
NHN Cloud	NVIDIA B200 (Blackwell)	2 nodes × 8 GPUs	16 GPUs — next-gen Blackwell bring-up, FP8/BF16 mixed-precision training
AWS	NVIDIA H200 SXM (141 GB)	2 nodes × 8 GPUs	16 GPUs — EFA/RDMA multi-node fabric, HBM3e-enabled long-context training

- **Parallelism & memory.** Trained models beyond single-node memory with DeepSpeed ZeRO-3, PyTorch FSDP, and Megatron tensor / pipeline parallelism, combined with activation checkpointing, Liger kernels, `torch.compile`, and FlashAttention-2/3 under BF16 / FP8 mixed precision.
- **Workloads.** Full-parameter and LoRA fine-tuning of **7B–30B** LLMs via **ms-swift** and **LLaMA-Factory**; VLM / Video-LLM multimodal training (vision encoder + projector + LLM stages); and **RLVR** / **policy optimization** (GRPO, PPO) on **verl**, **NeMo-RL**, and **slime** with vLLM-backed distributed rollouts.
- **Cluster stack.** SLURM scheduling with **Ray** orchestration; **Data-Juicer** / **NeMo Curator** for data curation; **lm-evaluation-harness**, **OpenCompass**, **EvalScope** for evaluation; **Megatron-Bridge** and **checkpoint-engine** for weight conversion across training–inference formats; **W&B** for tracking.

Heat-GRPO — Heat-Kernel Policy Optimization for Multi-Rubric RL

ICLR 2027 (under review)

- Built a **multi-rubric RL (RLAIF) training pipeline** in **verl** (FSDP + async vLLM rollouts) with a served **Qwen3-32B LLM-judge** returning per-rubric YES/NO log-prob margins as rewards — 8 rollouts per prompt, up to 16 signed rubrics per example.
- Implemented and trained **GRPO**, **GDPO**, **GD²PO**, and my **Heat-GRPO** (Gaussian-CDF scoring of judge margins before aggregation) on **6 policies** (Llama 1B–8B, Qwen3 1.7B–8B) × **4 benchmarks** (RaR-Science/Medicine, BeaverTails, HealthBench) × 3 seeds on H100s — **288 full RL runs**.
- Heat-GRPO achieved the best mean in **23/24** conditions (up to **+25%**), held up under **3 additional judges**, and cut advantage-computation cost by **~39%** vs. GDPO / GD²PO.

What Gets Deployed? — Deployment Accounting for Stateful Agents

ICLR 2027 (under review)

- Trained **LLM agents with multi-turn RL** in interactive environments (**ALFWorld**, **WebShop**) using **verl-agent** + **Ray**: Qwen2.5-7B, full-parameter FSDP on 4×H100, 16 sandboxed environment rollouts per task, 50-step action horizons.
- Implemented and compared **10 agentic RL credit-assignment variants** — GRPO, GiGPO (step-level credit), PKPO-style pass@k, leave-one-out multi-sample utilities, ternary abstention rewards, and a commit-aware credit — plus exploratory AppWorld runs (Qwen3.5-9B, LoRA).
- Built a **deployment-accounting** evaluation (commit / wrong-commit / abstain over sampled executions):

trained policies reached **72.5% / 71.3%** committed success on seen / unseen rooms vs. 62.9% / 60.5% for GRPO, with outcome selection adding **9–13 points**.

Subliminal RL — Tracing a Trait from Emission to Absorption

Nature (under review)

- Asked whether a **trait travels from a reward source into the policy through the reward signal alone**, when nothing in the generated content refers to it — decomposed into three independently breakable stages: **emission**, **transmission** through a semantically empty carrier, and **absorption** (the policy acts on it on held-out items). Where the chain breaks *is* the explanation.
- Built the system that makes this traceable: **prompt-level vs. weight-level** trait loading, carriers of **zero propositional content** (nonsense syllables, punctuation), and a trait axis from simple preferences up to **truthfulness** — with trait-ablated controls, disjoint readout items, and **decision rules fixed before seeing data**.
- Added an **observer axis** separating *semantic readability* from *statistical distinguishability* — yielding not one demonstration but a **map of the routes a trait takes through RL and where each breaks**.

Revisiting Continual Learning: A Three-Interface Diagnosis of Stability and Plasticity

NeurIPS 2026 – Main

- Showed that **BWT and FWT are interface-dependent**: measured through the sequential head, a fresh probe, or a classifier-free backbone diagnostic, the head suggests catastrophic forgetting, the probe near-complete recoverability, and the backbone only **partial representation drift**.
- Found a **backbone-level stability–plasticity trade-off** that is negative in **all 150** clustering configurations (mean $r = -0.74$) while head and probe correlations stay mixed in sign — across **7 datasets**, encoder and decoder backbones (BERT, RoBERTa, Pythia, Llama, Qwen up to 8B), and 8+ IL methods.
- Introduced **Just LM-Head Tuning (JLT)**, a head-only diagnostic intervention on the frozen incrementally trained backbone that recovers **+53 to +64 points** of accuracy (e.g., $40.7 \rightarrow 95.6$, near the 96.4 joint-training reference) — recasting much of “catastrophic forgetting” as **head–backbone mismatch**.

CLAWS — Creativity Detection for LLM-Generated Solutions

NeurIPS 2025 – Main

 github.com/kkt94/CLAWS

- Proposed **CLAWS**, a white-box detector using attention across prompt sections vs. output to classify solutions as *Typical* / *Creative* / *Hallucinated* — the first three-way creativity assessment for reasoning, with no human evaluation in the loop.
- Validated on **5 7–8B math RL models** over **4,545 problems from 181 contests** (AJHSME, AMC, AIME), outperforming **5 white-box baselines** (Perplexity, Logit/Window Entropy, Hidden Score, Attention Score) on both creativity and hallucination detection.
- Ensured evaluation rigor by scoring every signal under **5 evaluation strategies** (threshold, prototype, MLP, XGBoost, TabM) and reporting macro-averaged metrics to correct for the severe imbalance of the Creative and Hallucinated minority classes.

Thinking Diffusion — Visual-Grounded Reasoning for Diffusion MLLMs

CVPR 2026 – Main

- Conducted the **first quantitative analysis of CoT reasoning in dMLLMs**, identifying two failure modes: *Early Answer Generation* (model commits to the answer before reasoning) and *weak early-step visual grounding*.
- Proposed two training-free inference-time methods — **Position & Step Penalty (PSP)** and **Visual Reasoning Guidance (VRG)** — that delay premature answers and amplify visual conditioning via classifier-free guidance.
- Across LaViDa and MMaDA on *M3CoT*, *MMBench*, *ScienceQA-IMG*, *V* Bench*: **up to +7.5%** accuracy and **>3× faster** inference vs. baselines using 4× more diffusion steps. Plug-and-play across dMLLMs with no retraining.

ST-Veto — Spatio-Temporal Token Veto for Diffusion MLLMs

ICML 2026 – Main

- Designed a training-free, decoding-time **veto-and-swap mechanism** that exploits a property unique to dMLLMs: visibility of all tokens across diffusion steps.
- Vetoes (i) *temporally unstable tokens* via second-order Taylor prediction of confidence dynamics and (ii) *weakly-grounded tokens* via image-attention mass, then swaps them for safer candidates before unmasking.
- On LaViDa-reason and MMaDA-MixCoT across *M3CoT*, *MMBench*, *ScienceQA-IMG*, *V* Bench* under long generation lengths: **up to +9%** accuracy over baseline decoding and prior VLM reasoning methods, with **zero extra training or generation cost**.

- Proposed a **two-expert residual projector** (Local Expert + State Expert, both Gated DeltaNet) that augments any frozen Video-LLM’s base projector to explicitly model inter-frame motion and world dynamics.
- Introduced **JEPA-style auxiliary objectives** — local correspondence + masked state reconstruction — to specialize each expert without touching the vision encoder or LLM.
- Trained only an **~80K-parameter** module for a **single epoch on 70K public samples**, yielding **up to ~7%** gains across **7 Video-LLMs** (PLM, VideoLLaMA3, InternVL3 families) on **5 benchmarks** (MVBench, TempCompass, NExT-QA, WorldBench, Perception Test).

Education

Hanyang University, Ph.D. in Computer Science

Sep. 2023 – Present

Advisor: Prof. Yong Suk Choi | Seoul, Korea

Hanyang University, M.S. in Computer Science

Sep. 2021 – Aug. 2023

Advisor: Prof. Yong Suk Choi | Seoul, Korea

Kookmin University, B.S. in Automobile and IT Convergence

Mar. 2015 – Aug. 2019

Seoul, Korea

Research Interests

- LLM / VLM (Video-LLM) Post-training
- Interpretability, Alignment & Safety
- Agentic Reinforcement Learning; stateful agents, outcome-aware optimization
- Reinforcement Learning with Verifiable Rewards (RLVR); policy optimization for reasoning
- Video-LLMs, World Models, Vision-Language-Action models
- Continual / Incremental Learning (stability-plasticity)
- Diffusion Large/Multimodal Language Models (dLLMs, dMLLMs); visual-grounded reasoning

Publications (Journals & Conferences)

Bold + underline = author; * = first author.

1. **Keuntae Kim***, Yong Suk Choi. “CFP-AL: Combining Model Features and Prediction for Active Learning in Sentence Classification.” *Applied Sciences*, 2025. **PUBLISHED**
2. **Keuntae Kim***, Eunhye Jeong, Sehyeon Lee, Seohee Yoon, Yong Suk Choi. “CLAWS: Creativity detection for LLM-generated solutions using Attention Window of Sections.” *NeurIPS 2025 – Main*. **PUBLISHED**
3. **Keuntae Kim***, Mingyu Kang, Yong Suk Choi. “Thinking Diffusion: Penalize and Guide Visual-Grounded Reasoning in Diffusion Multimodal Language Models.” *CVPR 2026 – Main*. **PUBLISHED**
4. **Keuntae Kim***, Beomseok Lee, Hyunwoo Kim, Yong Suk Choi. “ST-Veto: Spatio-Temporal Token Veto for Diffusion MLLMs via Taylor Prediction and Visual Grounding.” *ICML 2026 – Main*. **PUBLISHED**
5. **Keuntae Kim***, Eunhye Jeong, Yong Suk Choi. “CWM: Controllable White-Box Meta-Prompting for Adaptive Retrieval-Augmented Generation and Reasoning Ability.” *Findings of EMNLP 2026*. **PUBLISHED**
6. **Keuntae Kim***, Beomseok Lee, Hyunwoo Kim, Yong Suk Choi. “Revisiting Continual Learning: A Three-Interface Diagnosis of Stability and Plasticity.” *NeurIPS 2026 – Main*. **PUBLISHED**
7. **Keuntae Kim***, Yong Suk Choi. “Two Clocks in Diffusion MLLMs: Answer Stabilization and Rationale Writing.” *NeurIPS 2026 – Workshop (BeNT)*. **PUBLISHED**
8. **Keuntae Kim***, Beomseok Lee, Hyunwoo Kim, Yong Suk Choi. “Heat-GRPO: Heat-Kernel Group Relative Policy Optimization for Multi-Rubric Reinforcement Learning.” *ICLR 2027 (under review)*. **UNDER REVIEW**
9. **Keuntae Kim***, JungHyouon Kim, Dongjun Kim, HyeongSeok Kang, Yong Suk Choi. “Projector-JEPA: A Gated DeltaNet Two-Expert Residual Projector Bridging Video-LLMs to World-Dynamics Understanding.” *ICLR 2027 (under review)*. **UNDER REVIEW**
10. **Keuntae Kim***, Eunhye Jeong, Sehyeon Lee, Seohee Yoon, Yong Suk Choi. “Can RLVR Models Think More Creatively? A Study of Overconfidence and High Entropy Segment.” *ICLR 2027 (under review)*. **UNDER REVIEW**
11. **Keuntae Kim***, Yong Suk Choi. “Subliminal Reinforcement Learning: Tracing a Trait from Emission to

Absorption.” *Nature* (under review). **UNDER REVIEW**

12. Beomseok Lee*, **Keuntae Kim***, Hyunwoo Kim, Yong Suk Choi. “What Gets Deployed? Deployment Accounting for Stateful Agents.” *ICLR 2027* (under review). **UNDER REVIEW**
13. JungHyoun Kim, **Keuntae Kim**, HyeongSeok Kang, Dongjun Kim, Yong Suk Choi. “Fourier-World: Temporal Fourier States for Visual World Models.” *ICLR 2027* (under review). **UNDER REVIEW**
14. Dongjun Kim, **Keuntae Kim**, JungHyoun Kim, HyeongSeok Kang, Yong Suk Choi. “Complete the Scene, Cover the Video: Recurrent Token Compression for Video LLMs.” *ICLR 2027* (under review). **UNDER REVIEW**

Research Grants & Funded Projects

- **Metacognitive AGI Framework and its Applications** — IITP. *Lead Researcher* (Jul. 2025 – Present).
- **Metacognitive Generative AI Framework for Mitigating Hallucination in LLMs** — NRF. *Lead Researcher* (Feb. 2025 – Present).
- **Big Data-based Real-time Monitoring and Social Simulation System** — NRF. *Lead Researcher, Group 3* (Sep. 2021 – Feb. 2025).
- **High-performance Deep Learning via Cross-connection of Autoencoders and Decoders** — NRF. *Researcher* (Sep. 2021 – Feb. 2024).
- **Natural Language Data Refinement for Large-scale LM Construction** — KT Corp. *Researcher* (Sep. 2021 – Jul. 2022).

Patents (Korea)

- “Fusion of Sparse-Dense and Global Attention for Video Action Recognition,” KR 10-2024-0132801.
- “Device and Method for Correcting Toxic Words in Sentence,” KR 10-2024-0143175.
- “Method and Device for Displaying Emotion Map Animation,” KR 10-2024-0152245.
- “Method for Analysing Emotion Based on SNS Data,” KR 10-2023-0193546.

Technical Skills

- **Training** — PyTorch, DeepSpeed (ZeRO-2/3), PyTorch FSDP, Megatron-LM, ms-swift, LLaMA-Factory, HF Accelerate, PEFT/LoRA, Liger kernels, bitsandbytes
- **RL & post-training** — verl, verl-agent, NeMo-RL, slime, TRL; GRPO / PPO / RLVR / multi-turn agentic RL with vLLM rollouts
- **Distributed & HPC** — multi-node training (up to 32×H100 / 4 nodes), SLURM, Ray, NCCL over InfiniBand / RoCE / AWS EFA, torchrun, tensor / pipeline parallelism, FlashAttention-2/3, BF16 & FP8, Megatron-Bridge / checkpoint-engine
- **Data & evaluation** — Data-Juicer, NeMo Curator, lm-evaluation-harness, OpenCompass, EvalScope
- **GPU platforms** — NVIDIA H100 / H200 / B200 (Blackwell), A100; Samsung SDS Cloud, NHN Cloud, AWS
- **Serving & inference** — vLLM, SGLang, TensorRT-LLM, HF TGI, ONNX Runtime, llama.cpp
- **Languages** — Python, C/C++, CUDA, JavaScript, TypeScript, Java, SQL
- **Web & backend** — Flask, FastAPI, Django, Node.js, React, REST APIs, WebSocket
- **Infra** — Linux, Docker, Git, W&B, MLflow, TensorBoard, NVIDIA Nsight / PyTorch Profiler

Awards & Honors

- **NeurIPS 2025 Scholar Award**
- **BK21 FOUR Scholarship** — Korean government-funded research fellowship, 2021 – Present

Academic Service & Activities

Conference Reviewer

- *Machine Learning* — **NeurIPS 2026** | **ICLR 2027** | **ICML 2026**
- *Computer Vision* — **CVPR 2026** | **ECCV 2026**
- *Natural Language Processing* — **ACL Rolling Review 2026.01** – Present

Workshop Presentation

- **NeurIPS 2026** BeNTo Workshop — *Two Clocks in Diffusion MLLMs: Answer Stabilization and Rationale Writing*